Informatics II, Spring 2026, Exercise 5

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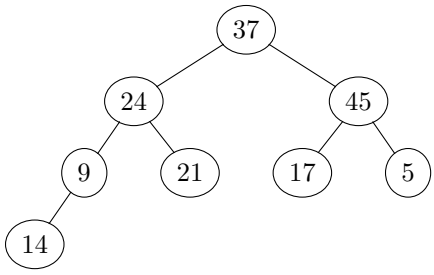
Task 1: Heapsort Algorithm

Given the array [37, 24, 45, 9, 21, 17, 5, 14], sort it in ascending order using Heapsort algorithm. Precisely explain every step performed. You can start from making max-heap from initially binary tree shown below. Show every node exchange that occurs.

Input array

37	24	45	9	21	17	5	14

Initial binary tree



Task 2: Heap and Heapsort

The d-ary heap is a generalization of the binary heap in which the nodes of the tree have **d** children instead of 2.

2.1 Give a mathematical formula that determines the indices of all child nodes of a node located at index n in a d-ary heap.

Write a C program by downloading the template from OLAT, then complete the code by implementing the specified functions (You can always create more helper functions).
Your program should first convert the given array into a max `d-ary` heap and print it in the standard output using the requested format. Then it should sort the heap array of integers in *ascending* order using `d-ary` heap sort and print the sorted array.
The functions to implement are:

2.2 *void printArray(int A[], int n)* It prints a given array to the console.

2.3 *void buildMaxHeap(int A[], int n, int d)* Where A is the array to be converted, n is the real size of the array A and d is the maximum number of child each node can have in the heap.

2.4 *void printHeap(int A[], int n, int d)* It prints the created max-heap to the console in the format `graph g {` (all the edges in the form `NodeA - NodeB`) `}`, where each edge should be printed in a separate line. The ordering of the edges is not relevant.

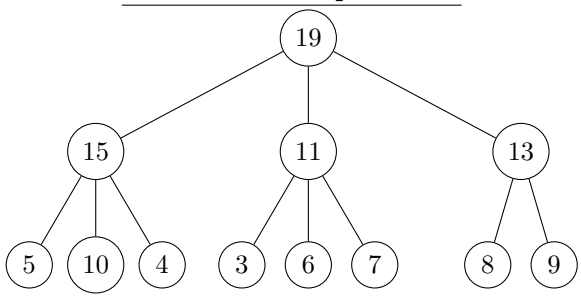
2.5 *void heapSort(int A[], int n, int d)* It sorts the array A in ascending order.

For example, given the array `A = [6, 4, 3, 9, 5, 10, 15, 19, 11, 7, 8, 13]` your program should produce the max-heap `[19, 15, 11, 13, 5, 10, 4, 3, 6, 7, 8, 9]` and print it to the console in the form illustrated on the left figure. Print the content of array A after calling *heapSort* on it. The alternative output is used to check the correctness of your implementation, it does not have to be part of your solution.

Output Form

```
graph g {
  19 -- 15
  19 -- 11
  19 -- 13
  15 -- 5
  15 -- 10
  15 -- 4
  11 -- 3
  11 -- 6
  11 -- 7
  13 -- 8
  13 -- 9
}
```

Alternative Output Form



Test your program with the array `[4, 3, 2, 5, 6, 7, 8, 9, 12, 1]`.

Note: You can copy the output of your program and use it on `http://www.webgraphviz.com/` to view your max-heap in the form of a binary tree, as illustrated on the right figure.

Task 3: Quicksort

In a $dual - pivot$ quicksort implementation, an array $A[low...high]$ is sorted based on two pivots $p_1 = A[low]$ and $p_2 = A[high]$. The pivot p_1 must be less than p_2, otherwise they have to be swapped. Here low is the left end of the array A and $high$ is the right end of the array A.

The partitioning process is the core of the $dual - pivot$ quicksort. We begin partitioning the array in three proper partition I, II or III, where:

- **Partition I:** In the first partition all elements will be less than the left pivot p_1.
- **Partition II:** In the second partition all elements will be greater or equal to the left pivot p_1 and also will be less than the right pivot p_2.

- **<u>Partition III:</u>** In the third partition all elements greater than or equal to the right pivot p_2.

After the partitioning, the two pivots are placed in their proper and final positions as illustrated in Figure 1 and the process is repeated recursively on each of the three partitions.

I		**II**		**III**
$< p_1$	$\mathbf{p_1}$	$>= p_1 \& < p_2$	$\mathbf{p_2}$	$>= p_2$

Figure 1: Quicksort with dual-pivot

Create a C program that implements $dual - pivot$ to sort an array A of integers in *increasing* order. Your program should also print the array before and after the sorting. For example, if the array A to be sorted is `[10, 7, 3, 15, 6, 2, 5, 1, 17, 8]`. your program should print the following in the standard output:

```
Before:  [ 10 7 3 15 6 2 5 1 17 8 ]
After:  [ 1 2 3 5 6 7 8 10 15 17 ]
```

You can find the template for your code on OLAT. You also might find helpful using some helper function you created in task 2 (Es. Print function) or creating new ones.